

A Review on Challenges and Applications of Digital Graphology

Ranjitha P¹, Akshatha Prabhu²

Department of Computer Science,

Amrita School of Arts and Sciences, Mysuru

Amrita Vishwa Vidyapeetham, India

rajithapramod123@gmail.com, akshathaprabhu06@gmail.com

Abstract—Handwriting is a process of using letters and symbols that follows a set of guidelines to communicate with others. It is a behavioral biometric that requires verification system to discriminate whether a signature is genuine or forged. In this paper a review of datasets, acquisition devices and preprocessing for offline and online systems followed by characters of handwriting like margins, slanting lines, upward writing, connected letters, speed and pressure points that deduce the writer's personality traits are proposed. The current state of the art has been discussed along with the accuracy. Support Vector Machines, Hidden Markov Model, Neural Networks, Dynamic Time Wrapping algorithms and Symbolic Representations are the promising approaches that has been mentioned in the later part of the paper. The final division in the paper highlights the applications of this system such as -Identifying heart and Parkinson's diseases, signature analysis and psychological assessment specifically on emotional state at critical events.

Keywords—Support Vector Machine, Kernel, Particle Swarm Optimization, Neural Network

I. INTRODUCTION

The various characteristics of an individual form a personality trait and it is through this that the individuals are being distinguished. These traits can be physical as in tall or short, slim or fat etc. Intelligence and creativity are the ability traits and communication, social gatherings form the social traits. It is cognitive, psychomotor, and biomechanical processes that directs the movement of hand, and fingers thus defining a person's behavior at a defined situation. Handwriting is a distinguishable characteristic that provides a valuable perception of an individual's behavior and measures the physiological state at the time of writing. It is a learned habit that has been a part of him/her defining expression of thought. It connotes intimate revelation of the inner processes of mental attitudes and emotions [1]. A powerful diagnostic tool that portrays sketch of person's mental anguish - emotional distress, strain, anxiety and restlessness, bridging emotions to the experience of a situation. Features in handwriting interpret unique and dynamic expression.

Graphology is a branch of science of character reading. It determines the personality profile and to interpret the individual's character. It identifies and determines the personality through patterns, curves and strokes [2][3]. It is the scientific method for recognizing, assessing and to understand writer's personality through the shapes and word patterns in the handwriting. Pressure, dimension, speed, form, direction, continuity and order are the seven elements of handwriting that are evaluated. It mimics and interprets the psychological, common sense and universal inferences. It

measures the inter-writer variations occurred due to variations in mood, muscular co-ordination, series of movements, frequency of writing, teaching methods, aesthetic inclination, and exposure to writings of others. Spatial features like spacing between and within characters, their strokes, and size, margin and base line relations, apparent pen pressure line spacing etc. Signatures in general are highly stylized motor program and due to limited samples often generate speculation on the authorship of a questioned signature [4].

The work is review on the various datasets – offline and online in Section II, Section III describes various parameters – Margins, Lines, Connected Components, Height and Size of Writing, Pressure and Speed. Section IV describes the various Methodology, Section V outline the applications of verification system. Conclusion and Future works is concentrated in Section VI. The input, approaches and output to the Signature Verification System and the Stages of the Verification System with input, preprocessing, feature extraction, Verification and Decision is depicted in “Fig.1”.

II. DATASETS

A. Offline datasets

Offline datasets consists of sample images of signatures taken on a plain white A4 paper that is scanned usually with a scanner, converting it to either JPEG or BMP format. These static systems use instruments to generate signals. It works on pixels of image generated by optical scanners where data to be operated is stored in two dimensional and space ordered domain. Standard GPDS database has 4000 contributors. Every individual have given 24 authentic and 30 forged signatures accounting to 96,000 genuine and 1,20,000 imitated signatures. MCYT-75 has 75 signers with 15 genuine and 15 forged signatures. There are 105 genuine, genuine pair and 225 genuine, forged pair. Persian signature dataset, UTSIG comprises 8280 images of 115 classes. Each class contributed by 6 forgers has 27 true signatures, and 42 skilled forgeries. CEDAR stores 2640 signatures of 55 contributors, each with 24 genuine samples. 24 forges samples are collected from each of 20 randomly chosen contributor. Available Online Databases are presented in “Fig.2”.

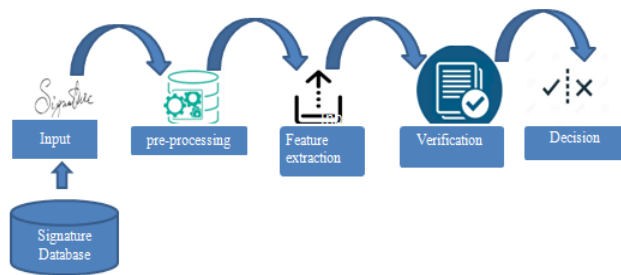


Fig.2: Stages of Handwritten Signature Verification System

B. Online datasets

Online datasets have data input with digitizing pen, smart pens, pressure sensitive tablet and stylus. These devices analyses pen trajectory that record co-ordinate values x, y along with pressure, acceleration, velocity etc. . . Log of these co-ordinates are conditioned on many factors - pen movement, time series, etc. These instantaneous data inputs may be stored in files for further analysis. On-line data consist of a time ordered sequence of coordinate points, the writing trace is known as each character is separable by time difference [5]. The system rejects the test signatures that compare its features against reference signatures based on the dissimilarity defined below a specific threshold. The method adopted shows better accuracy than offline verification [6]. Online verification is based on two principal approaches ; feature based and parameter based .In the former type , dynamic attributes like velocity , pressure , acceleration are represented as time functions whereas the latter is compact representation considering local and global parameters that represent to whole signature or critical points in signature[7][8][9].In feature based if low level feature extraction is performed then two non-linearly distorted functions are to be compared and in parameter approach the efficient parameter needs to be defined. The data acquisition step is crucial that depends on the quality of signals as it effects the preprocessing time. The parameters of the pen tip are referenced as functions of parameters and hence design of data acquisition tool is critical to capture inputs. Many researchers have designed and suggested instrumented pen and hardware to 4 measure calligraphic characters. They measure acceleration [10], force exerted and angle of the pen [11], hardware like digitizer [12][13] .Also , digitizing pen has been introduced [14] that measures the intensity of movement of the hand during the writing process by documenting the data of various measurements - the pressure , slant and gradient of the pen.

III. PARAMETERS

A. Margin

Margins are frames of a written text. It is one of the first peculiarities to be noticed that the writer has left on each page, representing bygone and prospective events. Left margins symbolize past and bond with family of origin while the right margins speak submission to pressure, anguish and fear of the future. Small, normal and big left margins are 10%, 15% and 25% of the width of the sheet of paper respectively while small and big right margins are 8% and 12% of the width of paper.

B. Lines

Maintaining handwriting on straight line – without being assisted by a line below – is, a sign of good health and sufficient physical strength. These directions relate to will, change in mood and emotional stability of the writer. The writing may be horizontal and straight on the line that reflects balanced state of mind;. The descriptions of various line types and associated description of human behavior is portrayed in “Fig.4”.

C. Height Or Size Of The Writing

The height/length of characters /letters contributes to emotional balance of the individual and skills at detailing. Measurements in dimension can be performed on middle area, upper extension, lower extension, capital letters and width. Balanced personality is expected to have moderate height.

D. The Slope Or Slant Of The Writing

Slant expresses writer’s greater or lesser predisposition for human contact, social interest and interpersonal skills. Graphologists have proved that the greater the slope or slant, the more sensitive, nervous and irritable the writer, often applicable to women with a psychological explanation that the female generally has ‘the stronger and more sensitive feeling and temperament. Slant of above 30o indicate irritability and intolerance [15].

E. Pressure

Pressure is the force applied that is applied perpendicular to the surface of the pen applied by the signer at the time of writing. Pen pressure varies from light to medium to heavy .The pen pressure displays thought flow and emotional intensity of the writer. The heavy pressure represents dynamic, self-assertive, and energetic and alertness as compared to light pressure shows passivity, non-receptive, lack of intensity and docile behavior. The moderate depicts balanced trait between light and heavy pressure.

F. Speed

The speed of writing is a more objective concept, it’s how fast a person writes that is how quickly or slowly the writer writes, inferring the intellectual characteristics 7 and the type of cognitive processes used to understand reality and resolve Fast handwriting is pretty rounded because angles slow down the writing, and usually vertical letters are written.

G. Connected lines

It is representative for logical or intuitive thinker. Connected writing have talent for introductions and bring people together, form conclusions and realizations with limited imagination and appreciable logical thinking. Connected and Not connected is depicted in “Fig.5”.

IV. METHODOLOGY

Preprocessing steps include background subtraction followed by binarization, noise removal and area open on signing videos [15].Methods like smoothing using Gaussian filter, elimination of time dependencies from signature uniformly resampling the signature with equidistant spacing are used [16].Preliminary steps can be achieved with

smoothing using cubic spline, translation, rotation and scale invariance with final step of pressure removal [17]. Isolated sample point removal is introduced. Euclidean measure is applied between this points to the remaining points. Time stamp removal removes the coordinates (x, y) amongst the set of points that have similar time stamp values [18]. Basic filter with rotation and normalization of co-ordinates are also followed [19].

Work has been progressed signature analysis using classifiers and models as illustrated in “Fig.6”. Verifications using SVMS have been carried by [20]. The work have defined a new kernel function using SVM. It is based on Longest Common Subsequences (LCSS) detection for time series for SVMs and their experimentation proves to be better than DTW based techniques. [21], based on generative Gaussian mixture models and discriminative polynomial classifiers which is compared against baseline systems.

Fisher and polynomial kernels have been simulated on YOHO database. [22] Have proposed one class support Vector machine for genuine signatures collected from CEDAR and GPDS datasets [23, 24]. The works have been implemented using time sequences and HMM on MCYT bimodal biometric database that employs more than 7000 signatures. [25] have proposed the log likelihood using HMM for a given handwritten signature. [26] have compared and observed relative small differences in performance of online and offline approaches where correctness rate for on-line is slightly better as against offline verification using discrete HMM with vector quantization. [27,28] have presented hybrid technique that uses global and local features. Local features using discrete features of HMM acquire the characteristics of signature production while global features capture spatial and temporal characteristics, together discriminating valid and forged attempts.

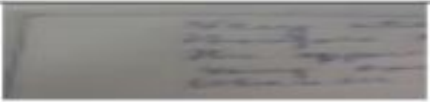
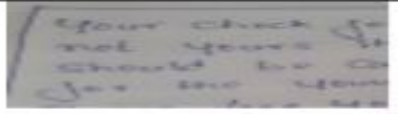
Margin Type	Description	Sample Image
No margin	Stingy person or miser who values great economy, personify grouchy and over carefulness to spend pennies, disrespect and selfishness	
Very wide margins	Liberty, generosity and extravagance, no conscious and at times dangerous impulsiveness with tendency to waste and spend.	
Very evenly kept margins	Calm and composure in making decisions, possessing evenness of mind and temper, love careful order and precision.	
Very uneven margins	Especially on the left of the page personify distortions in plans and actions, irregularity, unevenness carelessness, and changeable character - fickleness. Extremely uneven represents turmoil and great nervous unrest; sometimes leading to nervous breakdown and catration .	

Fig .3: Margin Type parameters with Sample images and Description

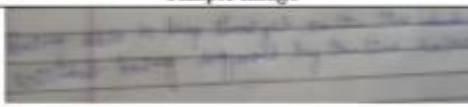
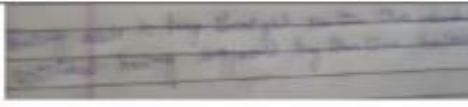
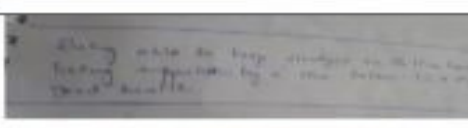
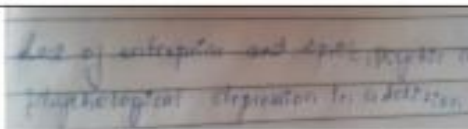
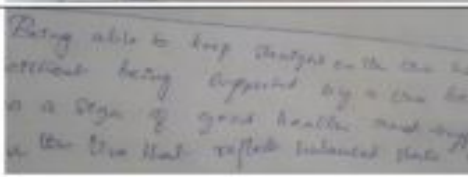
Line Type	Description	Sample Image
unable to write at all, except on lined paper	Set as weak, helpless with constant assistance and inability to comprehend.	
do not follow the printed lines but Write above, between or below them	Such writers are self-reliant , possessing individualistic nature , designs their own way in-life	
upward or rising lines	Show great diligence, confidence, enthusiasm in activities, projecting constructive and productive ideas	
Downward movements	Lack of enterprise and spirit , psychic and psychological depression in addition to anguish and weariness	
wavy-line writers	Accompanied by round, fine, thin or weak pen-strokes, indicates weakness, uncertainty and lack of independence. With regular even pen strokes, person is characterized as crafty, sharp witted with hypocrisy and diplomatic ability, in speech and manner.	

Fig. 4: Different Line in Handwriting with Illustrations and their associated meaning

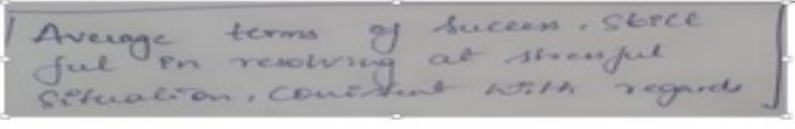
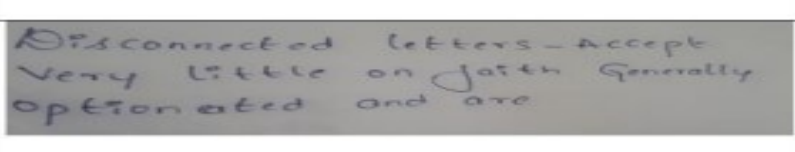
Connected	Careful, and prudent decision ,natural thinker and entrepreneurial	
Not connected	Accept very little on faith, Generally opinionated and are assertive. Hard to convince; tenacious, dogmatic on opinions.	

Fig. 5: Connected and Non Connected Lines that describe the individual behaviour

Structures. Overall error rate of 1.75% and 4.44 % for false rejection and false acceptance was observed respectively. Many techniques have been presented in Neural Networks. [29] Have used feed forward neural networks for 3 features namely – horizontal and vertical projection moments, upper and lower envelope of signature component on signature images.[30] have used Google Net architecture versions - Inception-v1 and Inception-v3 and have achieved validation accuracy of 83% and 75% respectively applied on the GPDS Synthetic Signature Database. [31, 32, and 33] Have experimented on four datasets, each one using CNN that has convolution and fully-connected layers, along with max-pooling, followed by Batch Normalization and ReLU non-linearity function after each learnable layer. [34] Have used 2 class Probabilistic Neural Network classifier whose parameters are tuned by PSO algorithm for verification process and the study was carried on SVC2004 and MCYT databases. Features and similarity coefficients of behavioral biometrics. [35] Have experimented their approach on (GPDS) database and introduced the applications using Particle Swarm Optimization which is a of stochastic optimization technique that can be used for training neural networks. 'NN with PSO algorithm is applied by [36] with acceptance and rejection rates being 75% and 25% for

Genuine signatures. Symbolic representation have been proposed by [37, 38, and 39] that is implemented two stages: The first stage is symbolic representation of signatures and second is symbolic matching used for verification and recognition.[40,41] Have introduced Stability Modulated Dynamic Time Warping (SM-DTW) algorithm that detects the stability regions that represent the most similar parts between two signatures. The summary of the classifier - SVM, HMM and Symbolic Representations with the algorithms and techniques used by the referenced papers have been tabulated as shown in Table 1. The Tablet and Digitizer have been represented with the letter (T) and (D) respectively under the column Database/Tablet/Digitizer. The Ref column in TABLE 1 is the referenced paper as given in the references section of the paper. The Error Rate(ER) and Equal Error Rate (EER) is measured in %. The summary of the classifier - Neural Networks (NN) and Dynamic Time Warping (DTW) with methods exploited by researchers in the referenced paper (as illustrated as Ref in Table) are shown in TABLE 2. Manual refers to the non-usage of any standard database by the authors implying that they have created a database of their own by collecting samples of signatures manually .The

Error Rate(ER) and Equal Error Rate (EER) is measured in %.

V. APPLICATIONS

Presence of a heart disease by analyzing Handwriting of an individual even before physical symptoms appear have been shown [19]. The Heart diseases is considered as one of the major causes of mortality and thus there is constant need for early prediction of heart disease. The work adopted uses

a k-Nearest Neighbor .It is applied on ten writing features. Prediction of low blood pressure and diabetes has achieved 75 % accuracy [20, 21, and 22]. Sentiments such as - Contempt, Anger, Disguise, Joy, Sad, Surprise, and Fear have been analyzed. . The work proposed uses decisive algorithm that takes input images .The method proposed have posed a challenge on data of students and teachers of different thoughts and varying moods .

Classifier/Method	Algorithm	Database /Tablet/ Digitizer	Ref	Error Rate or Equal Error Rate (ER/EER in %)
SVM	Longest Common Subsequences (LCSS)	SVC 2004	[20]	0.60 EER for skilled
	Curvelet Transform , Soft Thresholding	CEDAR, GPDS	[22]	16.92 EER for 4 genuine 15.95 EER for 8 genuine 15.07 EER for 12 genuine
	Geometric normalization ,Time derivatives , Signal normalization	MCYT	[23]	0.74 EER for skilled 0.05 EER for random
HMM	Structural user-dependent , Geometric normalization	MCYT	[24]	6.14 EER for skilled 2.09 EER for random
	Adaptive Thresholding , Linear Discriminant Analysis	Philips Advanced Interactive Display (PAID) (T)	[25]	1.0 - 1.9 EER for skilled
	Normalized Fourier coefficients , cubic smoothing B-splines , Viterbi algorithm , Segmental k-means iterative procedure	Murray Hill	[27]	2.5 EER
	Baum-Welch algorithm, forward and backward probabilities	DSD 703 (D)	[28]	4.44 ER for false acceptance 1.75 ER for false rejection
	Feature dependent threshold, mean standard deviation , intervalued feature vector	MCYT	[37]	5.35 EER
Symbolic Representation	filter based feature selection , Mean Absolute difference , Mean Median difference , Modified arithmetic & geometric ratio	MCYT	[38]	For writer dependent parameters 2.2 EER for 5 skilled writers 0.6 EER for 20 skilled writers

Fig 6: Summary of SVM, HMM and Symbolic Representation in Signature Verification System

[23] Propose a set of features for automatic signature verification. These features are evaluated on standard database consisting of 5,600 signatures using hidden Markov model applied specifically in the context of bank tuque processing. [24,25] Have carried out handwriting analysis to infer gender and age estimates and have stated limited information concerning the personality, age, gender and intent of the writer.

V. CONCLUSION

Large variety of methods have been proposed by many researchers for identification of forged and genuine signatures. Error rates in distinguishing them have reduced considerably due to advancement in neural networks and Deep Learning methods. The methods suggested by many authors such as SVM, HMM, PSO and DWT algorithms and symbolic representations lead to promising directions. Commonly used datasets for experimentation, acquisition devices and pre-processing have been briefed. Applications of verification systems for sentiment analysis, Parkinson's

disease, psychological assessment and emotional state has been mentioned. In the authors' opinion, researchers will continue to explore feature extraction, identification of the critical features in signature, classifiers particularly based on neural networks and Deep Learning approaches.

For future work, considerable amount of work needs to be proceeded on certain areas like Alzheimer's based signature traits, identification of gender; exploration of interrelationships between the different elements of mood and exploit this information to classify participants' diagnosis. Signatures have a vital role in banking system. Measures have to be taken to store the client's change in signature. . There is no standard device used specifically for online signature verification. Few authors develop their own devices as per specific applications as against offline signatures that work on scanned images. These devices are susceptible to noises, distortion and fluctuations during the acquisition. Validations on few test samples have to be made with the manual experts to verify the result obtained.

TABLE 2: Summary of Neural Network and DTW in Signature Verification System

Classifier/ Methods	Algorithm	Database/ Tablet or Digitizer	Ref	Error Rate or Equal Error Rate (ER/EER in %)
Neural Networks	Feed Forward Neural Net , Backpropagation , Connectionist Scheme , Thresholding	Handheld Scanner	[29]	3.33 ER for false acceptance 1.0 for false rejection
	Google Net - Inception-v1 , Inception-v3	GPDS Synthetic Signature	[30]	17 EER for Inception-v1 24 EER for Inception-v3
	Otsu algorithm , CNN - convolutions , maxpooling , dot products and Batch Normalization. Stochastic Gradient Descent, SigNet and SigNet-F (Features)	GPDS	[31]	1.72 (+0.15) EER on SigNet-F
		MCYT		2.87 (+0.42) EER on SigNet
		CEDAR		4.63 (+0.42) EER on SigNet-F
		PUC-PR		2.01(+0.43) EER on SigNet
	Probability Density Function , Feed forward Neural network with BP learning algorithm , KNN classifier	Manual	[32]	3.22 for False Positive
	Grid Segmentation , Cham code , Euclidian distance measure	Manual	[33]	10.33 for False Positive 4.41 for False Negative on Random Samples
		SVC2004	[34]	0.11for False acceptance 0.53 for False Rejection
	Hotelling's algorithm , Probabilistic Neural Network , Particle Swarm Optimization(PSO)	MCYT		0% for both false acceptance and rejection
	Global best PSO with Neural Network (PSO-NN)	(GPDS)	[35]	26.85 for False acceptance on skilled 17.33 for False rejection on forged
Dynamic Time Warping	Normalization, Thinning , equal size distribution and the square grid segmentation, PSO-NN	Manual	[36]	75 % acceptance rate 25% rejection rate
	(DTW , Fuzzy rules , Principal Component Analysis(PCA)	MCYT -100	[39]	4.88% average error rate
		BioSecure		3.33% average error rate
	Bresenham algorithm, weighted edit distance , distance normalization	MCYT -100	[40]	1.30% EER for skilled 3.09% EER for forged
		BioSecureID-SONOF		1.17% EER for skilled 1.45% for forged
	Maximum Length Vertical Projection (MLVP) , Minimum Length Horizontal Projection (MLHP) , DTW	Manual	[41]	2% EER

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